Calculating Aerodynamics Forces and Moments from Surface Pressure and Shear Stress Data

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Nomenclature

C_p	=	Pressure Coefficient
C_L	=	Lift Coefficient
C_M	=	Moment Coefficient
C_d	=	Drag Coefficient
c	=	Chord
z	=	Camber
α	=	Angle of Attack
a	=	lift Slope
M	=	Mach Number
m	=	Maximum Camber Value
p	=	Value of $\frac{x}{c}$ relative to m
$\frac{x}{c}$	=	Airfoil Location
x	=	Chord Line Distance
Re	=	Reynolds Number
$C_{m,\,c/4}$	=	Moment about quarter Chord
$C_{m,\,LE}$	=	Moment about Leading Edge
A_1	=	Lift Distribution Fourier Coefficient
A_2	=	Curvature Distribution Fourier Coefficient

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I. Introduction

The study of aerodynamic forces and moments acting on airfoils is fundamental to the design and optimization of aircraft wings and other lifting surfaces. In this project, we explore the aerodynamic performance of the Liebeck LA2573A airfoil, a high-lift design known for its efficiency in low-speed applications, using XFLR5, an open-source computational tool based on the XFOIL engine. By integrating surface pressure and shear stress data, this investigation aims to calculate key aerodynamic coefficients—lift coefficient (Cl), drag coefficient (Cd), and quarter-chord moment coefficient (Cm)—while assessing the influence of Reynolds number and angle of attack on airfoil behavior. XFLR5 provides a robust platform for simulating airflow over the airfoil, enabling the generation of pressure (Cp) and skin friction (Cf) distributions critical to our analysis. Working in a team of three, we developed an Excel spreadsheet to process XFLR5 outputs, perform numerical integrations, and visualize the results through graphs such as airfoil geometry, Cp distributions, and Cf variations. This project not only enhances our understanding of airfoil aerodynamics but also evaluates the practical application of computational tools in predicting performance across a range of operating conditions, specifically at a Reynolds number of 250,000 over angles of attack from -2° to 10°, and across a broader Reynolds number range from 250,000 to 1,000,000 at a fixed angle of attack of 4°. Through this effort, we aim to bridge theoretical principles with computational analysis, offering insights into the Liebeck LA2573A airfoil's response to varying flow regimes.

II. Procedure

We observed the airfoil, on how changing only the maximum camber, changing only the location of the maximum camber, and changing both the maximum camber and its locations from the leading edge effect the lift curves in inviscid and viscous flow, and when modelled using XFLR5. To start the calculations, the mean camber line was found. The equations

$$z = \frac{mc}{p^2}\left[2p\left(\frac{x}{c}\right) - \left(\frac{x}{c}\right)^2\right] \tag{1}$$

for the airfoils where $\frac{x}{c}$ was less than or equal to the chord location and

$$z = \frac{mc}{(1-p)^2}\left[(1 - 2p) + 2p\left(\frac{x}{c}\right) - \left(\frac{x}{c}\right)^2\right] \tag{2}$$

for all airfoils where $\frac{x}{c}$was greater than or equal to the chord location, and was used. From there, we replaced $\frac{x}{c}$ with $\frac{1-cos\theta}{2}$ to get both equations into the form of $\frac{dz}{dx}$. The bounds of the equation were also changed by setting $\frac{1-cos\theta}{2}$ equal to the chord location and solving for theta. From there, the angle of lift can be found by using the equation

$$\alpha_{L=0} = -\frac{1}{\pi}\int_0^{\theta}\left(\frac{dz}{dx}\right)(\cos\theta_0 - 1)d\theta_0 - \frac{1}{\pi}\int_{\theta}^{\pi}\left(\frac{dz}{dx}\right)(\cos\theta_0 - 1)d\theta_0 \qquad (3)$$

After that, the moment coefficient about the quarter chord can be found. To do this, the Fourier Coefficients need to be found by using the equations

$$A_1 = \frac{2}{\pi}\int_0^{\theta}\left(\frac{dz}{dx}\right)\cos\theta d\theta + \frac{2}{\pi}\int_{\theta}^{\pi}\left(\frac{dz}{dx}\right)\cos\theta d\theta \qquad (4)$$

And

$$A_2 = \frac{2}{\pi}\int_0^{\theta}\left(\frac{dz}{dx}\right)\cos 2\theta d\theta + \frac{2}{\pi}\int_{\theta}^{\pi}\left(\frac{dz}{dx}\right)\cos 2\theta d\theta \qquad (5)$$

Then those Fourier Coefficients can be used in the equation

$$c_{m,\frac{c}{4}} = \frac{\pi}{4}(A_2 - A_1) \qquad (6)$$

to solve for the moment above the quarter chord. For the coefficient of lift, we used the equation

$$C_l = \frac{L}{\frac{1}{2}\rho u^2 S} \qquad (7)$$

where this equation utilizes the airfoil surface area, flow speed and flow density. The equation

$$c_{m,LE} = -\left[\frac{Cl}{4} + \frac{\pi}{4}(A_1 - A_2)\right] \qquad (8)$$

can be used to find the moment coefficient about the leading edge. The center of pressure can be found using the equation

$$x_{cp} = \frac{c}{4}\left[1 + \frac{\pi}{Cl}(A_1 - A_2)\right] \qquad (9)$$

We also can use the trapezoidal rule. To determine our coefficients for drag and moment, we use the equation

$$\int_0^c Cp(x)dx \approx \sum_{n=1}^{N}\frac{Cp_{n-1} + Cp_n}{2}\Delta x_n \qquad (10)$$

Skin friction drag then can be determined by

$$C_f = \frac{\tau_\infty}{\frac{1}{2}\rho_\infty v_\infty^2} \tag{11}$$

Similarly, we then utilize the coefficient of pressure where the equation uses the static pressure coefficient ,freestream of fluid density, velocity, and static pressure. The equation

$$C_p = \frac{p-p_\infty}{\frac{1}{2}\rho_\infty v_\infty^2} \tag{12}$$

is used. To approximate over the entire span of the upper and lower surface of the airfoil Simpson's rule was used. Therefore we used the equation

$$\int_a^b f(x)dx \approx \frac{b-a}{6}[f(\alpha) + 4f(\frac{b-a}{6}) + f(b)] \tag{13}$$

The pitching moment coefficient is also determined by the Simpsons rule by using

$$C_m = \frac{M}{qS_c} = [C_p sin(\alpha) + C_f cos(\alpha)](x - 0.25)dx \tag{14}$$

Along with our coefficient of drag given by

$$C_d = \frac{2F_d}{\rho u^2 A} \tag{15}$$

where this equation utilizes the drag force, density of fluid and flow speed relative to the area.

Thin airfoil theory is then used to solve the coefficient of lift. The XLR5 software helps us to find the coefficient of lift for both viscid and inviscid flow. The angle of attack was calculated from a range for these simulations.Then XLR5 modeled the airfoil in a viscid flow using a range of -2 to 10 degrees. To run the program in XFLR5, we began by defining the Liebeck LA2573A airfoil. We did this by importing the airfoil coordinates into XFLR5 via the "Define an Airfoil" option under the "File" menu. Once the airfoil is imported, refine the paneling to ensure accurate calculations, particularly in the leading and trailing edges, to improve our resolution of pressure coefficient (C_p) and skin friction coefficient (C_f) distributions. Next, we set up an analysis by navigating to "Analysis" → "Define Analysis," where we will specify the Reynolds number (Re=250,000 and the range from 2.5E to 1.0E+6) and select the XFOIL solver. Choosing The angles of attack from −2 degrees to 10 degrees in 2-degree increments, ensuring the calculations align with the specified parameters.

Then run the analysis to compute lift coefficient (C_l), drag coefficient (C_d), and moment coefficient (C_m) for each angle of attack.We export the results into an Excel-compatible format by selecting "Export Polar" from the results window. To obtain the pressure coefficient (Cp) and friction coefficient (C_f) distributions, we select an angle of attack within the desired range, to visualize and extract the required data. Afterwards we export these distributions for different angles of attack to integrate into the Excel template.

In Excel, we input the airfoil coordinates into a designated sheet to plot the airfoil shape. Set up the "Calculation" sheet to link angle of attack inputs with their corresponding C_p and C_f values, enabling automated calculations of C_l, C_d, and C_m through numerical integration of pressure and shear stress data. Finally, we generate the required plots, including labeling and formatting them for clarity. This process ensures a structured workflow from XFLR5 simulations to Excel-based data processing and visualization.

To guarantee efficiency, my team member and I worked closely together on the project, allocating assignments according to our individual skills. To coordinate our efforts, exchange ideas, and resolve any issues, we kept in regular contact. We ensured a well-rounded approach by having one of us manage execution and implementation while the other concentrated on research and preparation. Throughout the process, we gave each other comments, improving our work to produce the greatest result. The project was successful because of our cooperation and joint effort.

III. Results

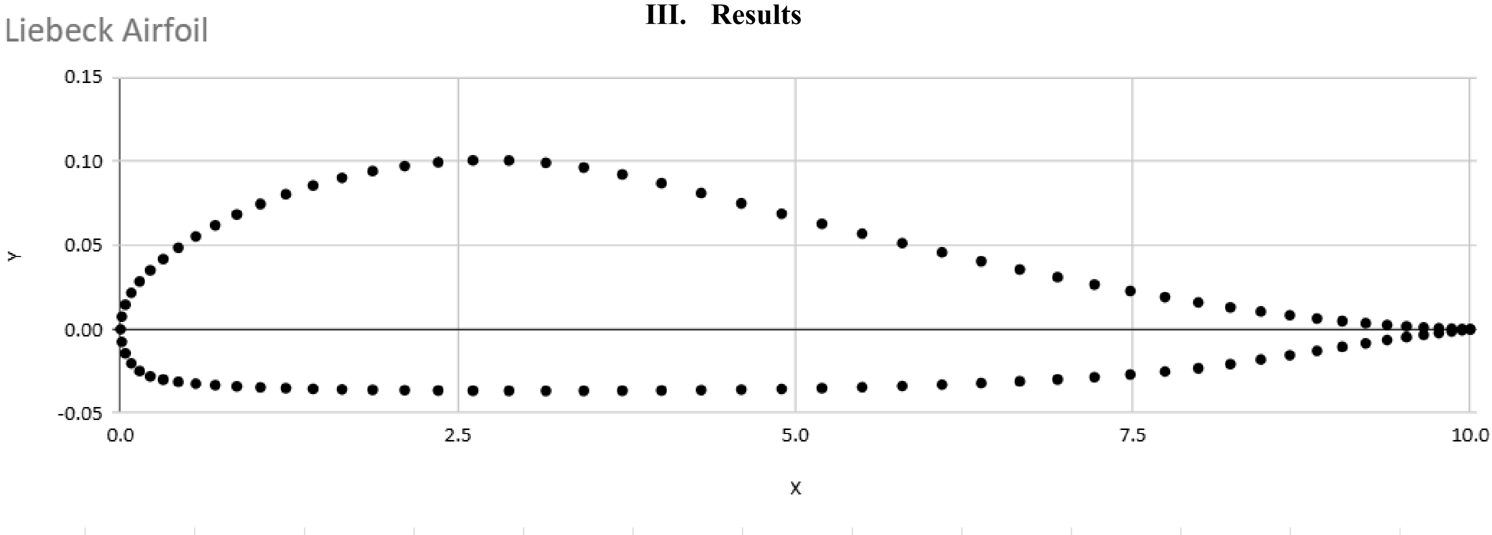


Fig. 1: Liebeck LA2573A Airfoil Shape

Figure 1 shows the Liebeck LA2573A airfoil shape, with X-axis representing chord-wise position and Y-axis showing thickness distribution. The black dots outline the airfoil's smooth upper surface, flatter lower surface, and tapered trailing edge, characteristic of high-lift airfoils.

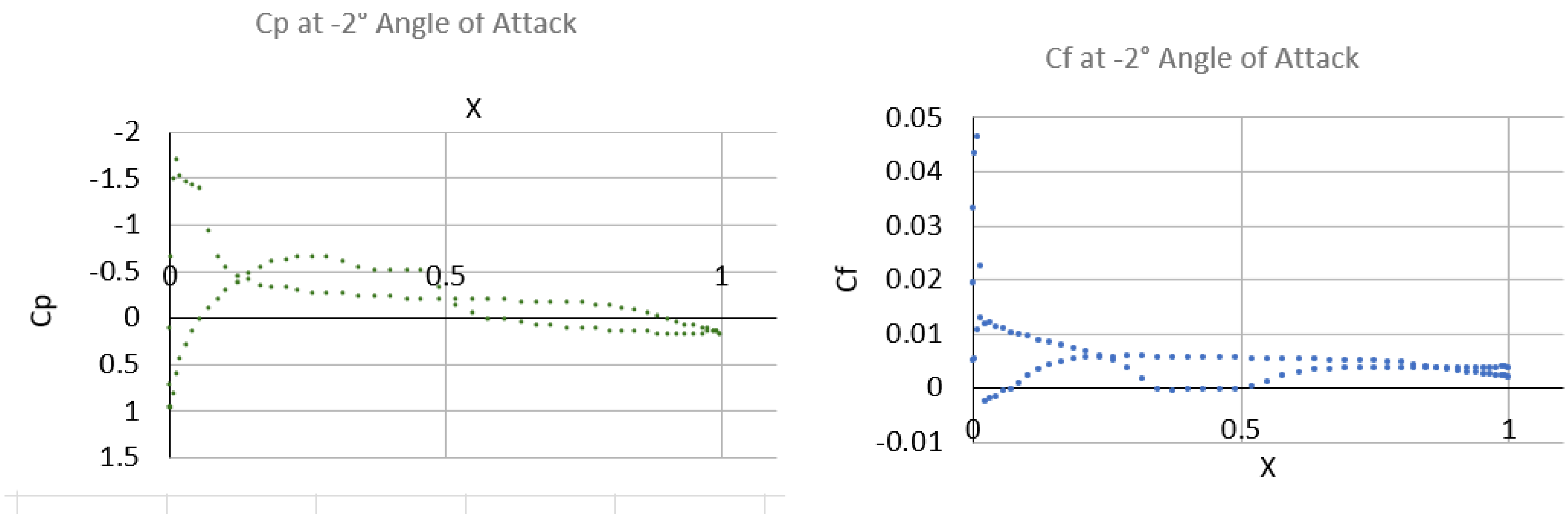


Fig. 2 & 3: Cp and Cf angle of attack at $\alpha = -2$ degrees

Figure 2 and 3 shows the pressure coefficient (Cp) and skin friction coefficient (Cf) distributions along the airfoil surface at an angle of attack of -2°. The left graph presents Cp vs. X, indicating pressure variations, while the right graph displays Cf vs. X, showing surface friction behavior. Both plots highlight aerodynamic characteristics at this negative angle of attack.

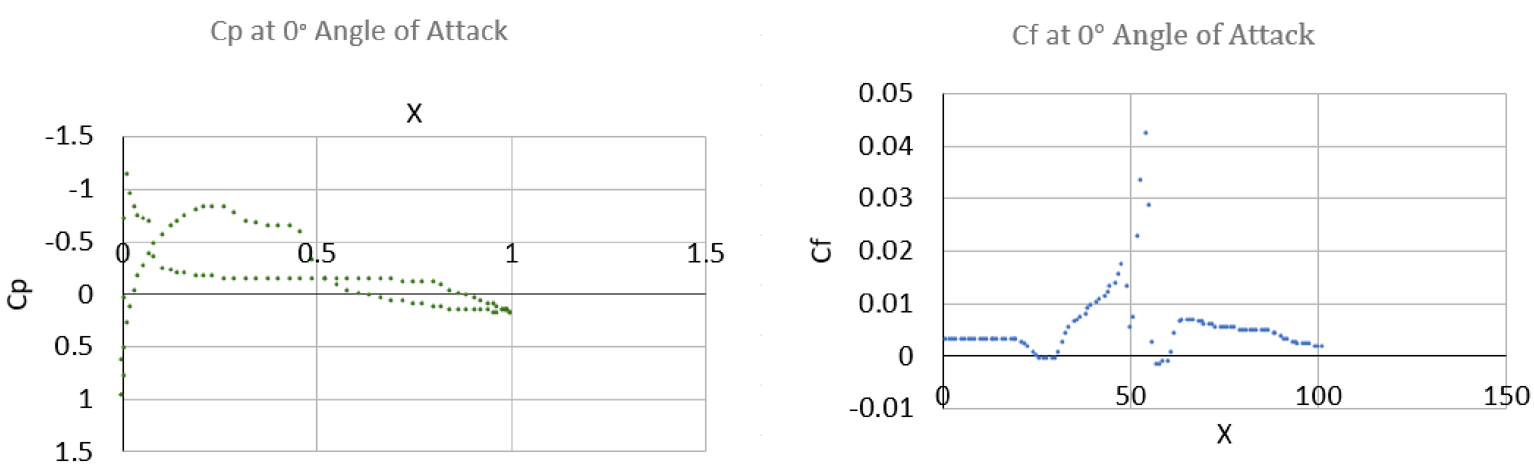
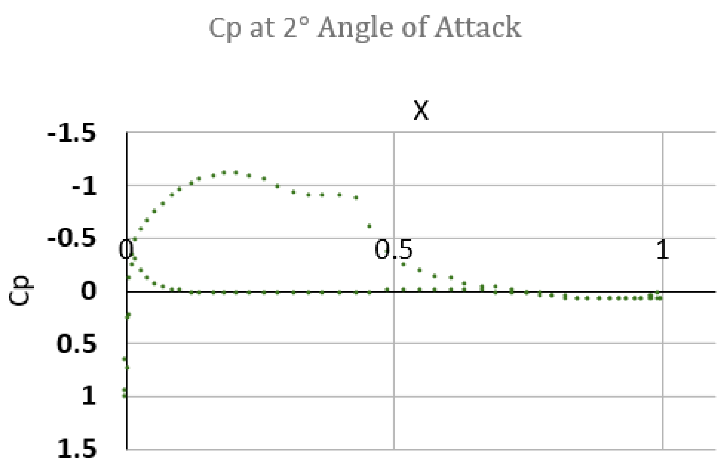


Fig. 4 & 5: Cp and Cf angle of attack at $\alpha = 0$ degrees

The graphs illustrate the pressure coefficient (Cp) and skin friction coefficient (Cf) distributions at 0° angle of attack. Figure 4 shows Cp vs. X, highlighting pressure variations along the airfoil surface, while the right Figure 5 presents Cf vs. X, depicting surface friction characteristics. These plots provide insight into the aerodynamic performance at this neutral angle.



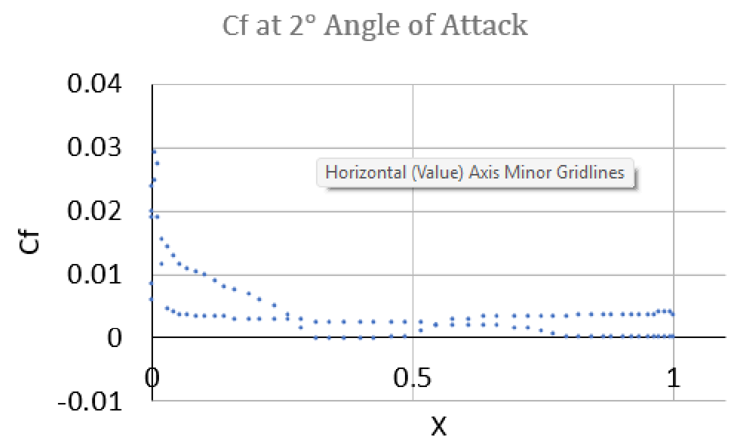
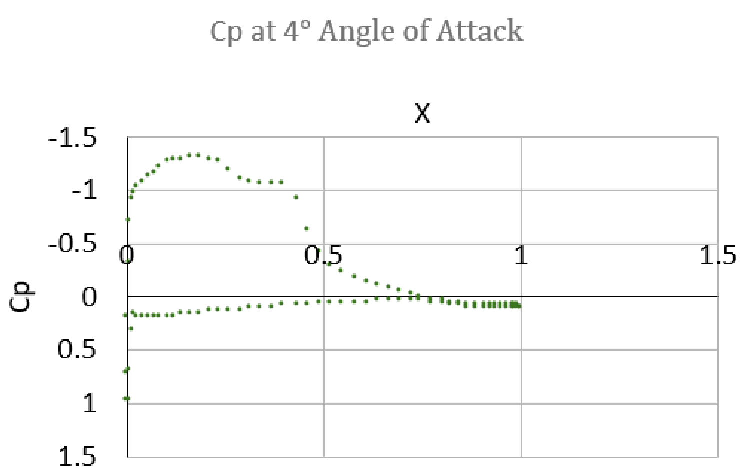


Fig. 6 & 7: Cp and Cf angle of attack at $\alpha = 2$ degrees

The graphs show the pressure coefficient (Cp) and skin friction coefficient (Cf) distributions at an angle of attack of 2°. Figure 6 illustrates Cp vs. X, depicting pressure variations along the airfoil, while Figure 7 presents Cf vs. X, highlighting surface friction behavior. These results reflect aerodynamic changes due to the increased angle of attack.



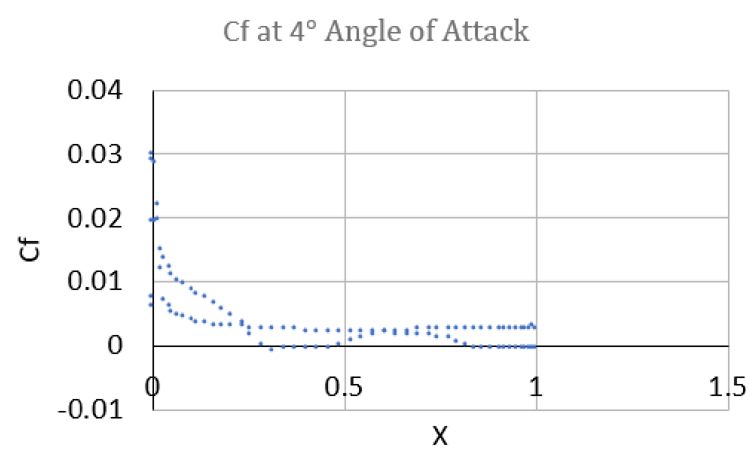
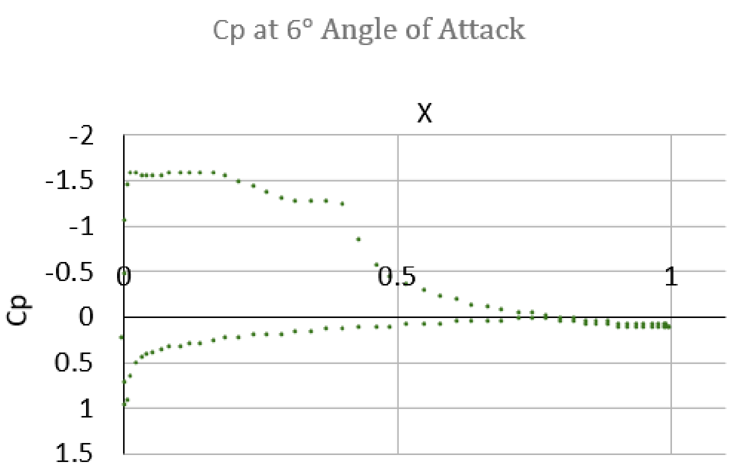


Fig. 8 & 9: Cp and Cf angle of attack at $\alpha = 4$ degrees

Figure 8 and 9 demonstrate the pressure coefficient (Cp) and skin friction coefficient (Cf) distributions at an angle of attack of 4°. Figure 8 illustrates Cp vs. X, indicating pressure variations along the airfoil, while Figure 9 presents Cf vs. X, showing surface friction behavior. These results highlight the aerodynamic effects of a higher angle of attack.



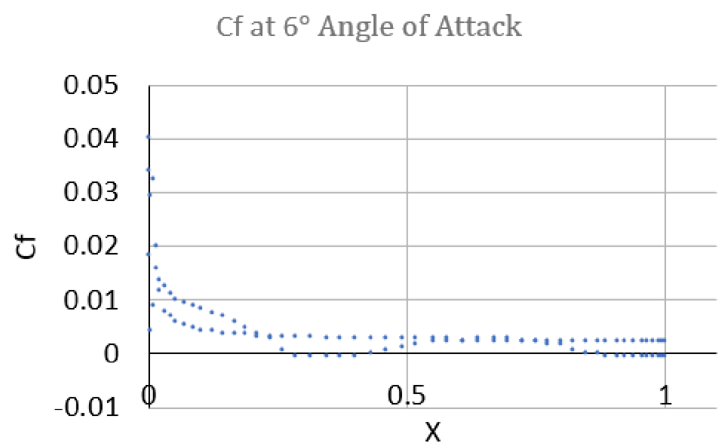
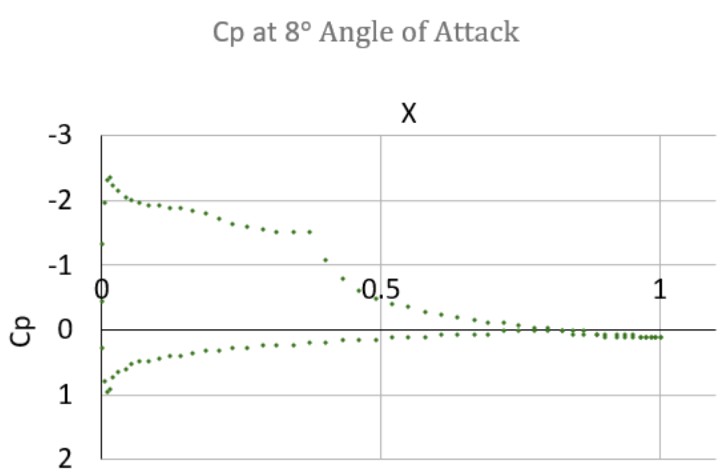


Fig. 10 & 11: Cp and Cf angle of attack at $\alpha = 6$ **degrees**

The graphs represent the pressure coefficient (Cp) and skin friction coefficient (Cf) distributions at an angle of attack of 6°. Figure 10 illustrates Cp vs. X, depicting pressure variations along the airfoil, while Figure 11 presents Cf vs. X, showing surface friction behavior. These results demonstrate the aerodynamic changes as the angle of attack increases.



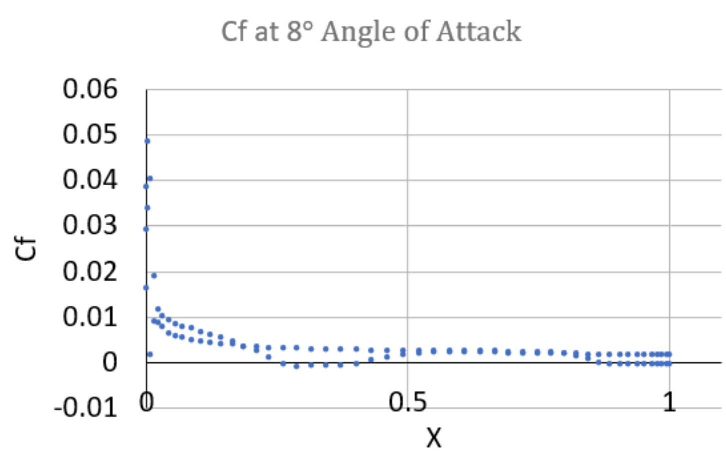
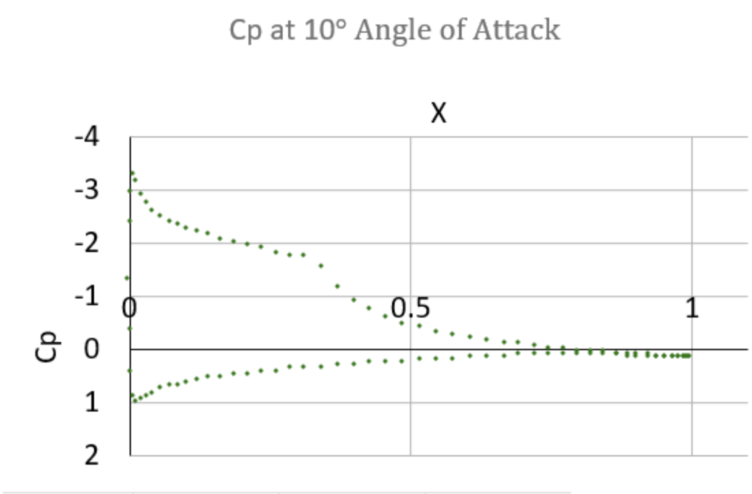


Fig. 12 & 13: Cp and Cf angle of attack at $\alpha = 8$ **degrees**

The graphs show the pressure coefficient (Cp) and skin friction coefficient (Cf) distributions at an angle of attack of 8°. Figure 12 illustrates Cp vs. X, showing pressure distribution along the airfoil. Figure 13 presents Cf vs. X, highlighting surface friction. These results demonstrate the aerodynamic effects as the angle of attack increases further.



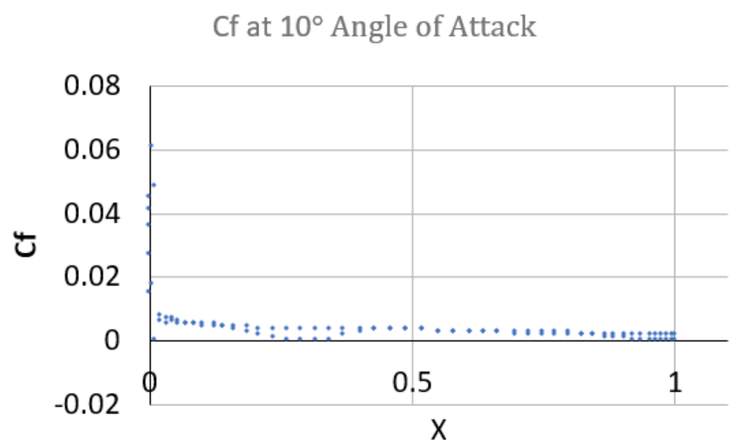


Fig. 14 & 15: Cp and Cf angle of attack at $\alpha = 10$ degrees

Figure 14 and 15 the pressure coefficient (Cp) and skin friction coefficient (Cf) distributions at an angle of attack of 10°. Figure 14 illustrates Cp vs. X, showing pressure variation along the airfoil, while. Figure 15 presents Cf vs. X, highlighting surface friction. The results indicate increased aerodynamic effects at this higher angle of attack.

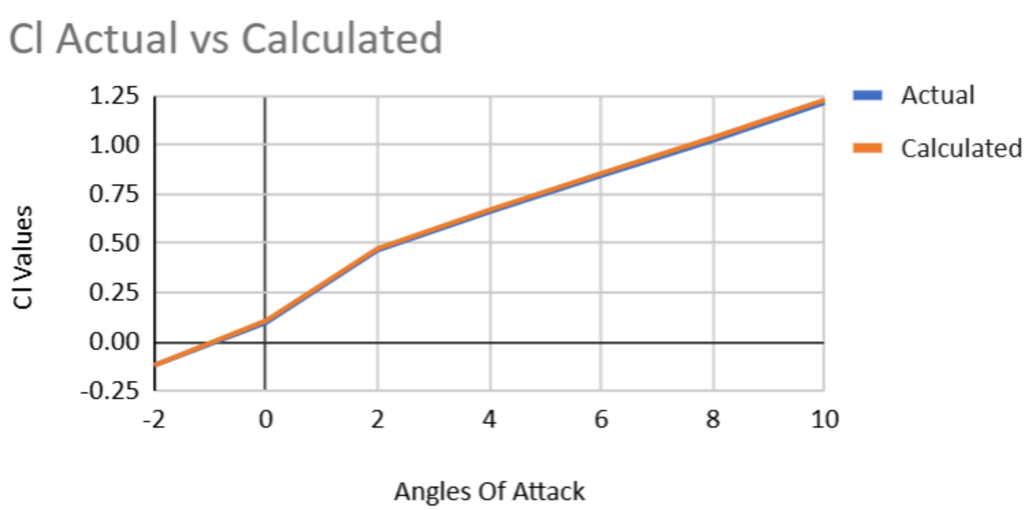


Fig. 16: Actual Cl vs Calculated Cl values

Figure 16 compares actual vs. calculated lift coefficient (Cl) values across different angles of attack. The blue line represents actual Cl values, while the orange line represents calculated Cl values. The close alignment of both curves suggests good agreement between the theoretical and experimental results, indicating an accurate aerodynamic model.

Fig. 17: Cl Percent error across the Angles of Attack

Figure 17 shows the percent error in lift coefficient (Cl) across different angles of attack. The highest error occurs at 0° angle of attack, reaching around 11%, while errors at other angles remain relatively low. This suggests that the calculation method is more accurate at higher angles of attack.

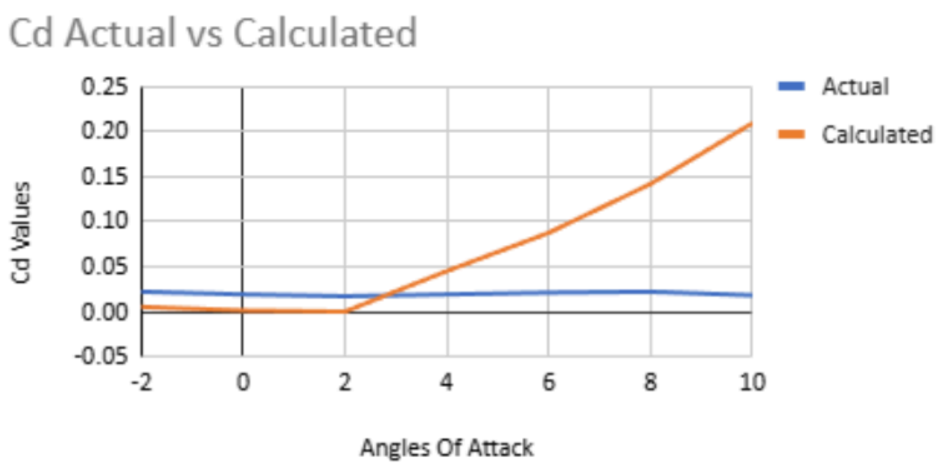


Fig. 18: Actual Cd vs calculated Cd values

Figure 18 compares actual vs. calculated drag coefficient (Cd) values across different angles of attack. The blue line represents actual Cd values, while the orange line represents calculated Cd values. The discrepancy increases at higher angles of attack, suggesting that the calculation method overestimates drag at larger angles.

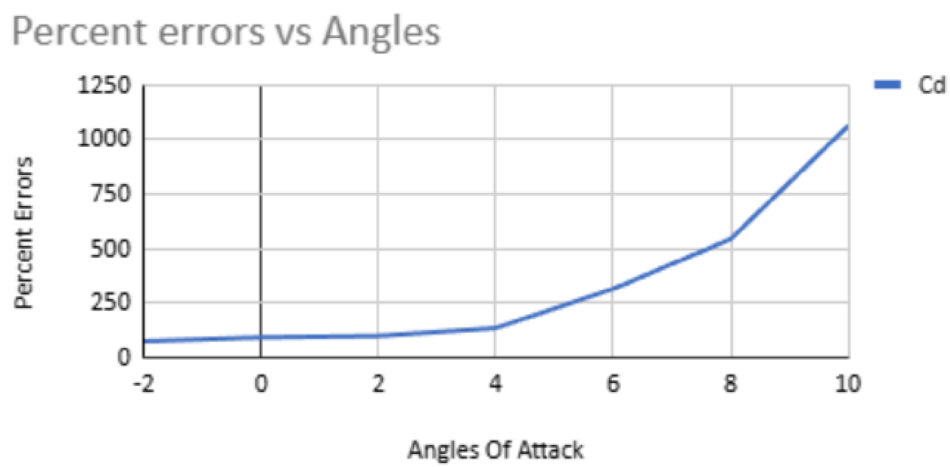


Fig. 19: Cd Percent error across the Angles of Attack

Figure 19 shows the percent error in drag coefficient (Cd) across different angles of attack. The error remains relatively low at small angles but increases significantly at higher angles, exceeding 1000% at 10°. This suggests that the calculation method struggles to accurately predict drag at larger angles of attack.

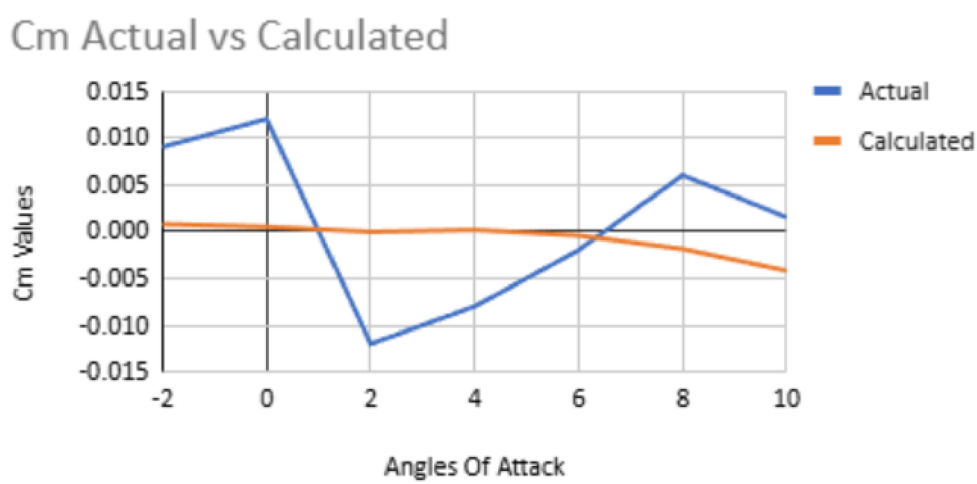


Fig. 20: Actual Cm vs calculated Cm values

Figure 20 compares actual vs. calculated moment coefficient (Cm) values across different angles of attack. The blue line represents actual Cm values, while the orange line represents calculated Cm values. The discrepancies between the two indicate variations in predicted vs. measured pitching moments, suggesting potential inaccuracies in the computational model.

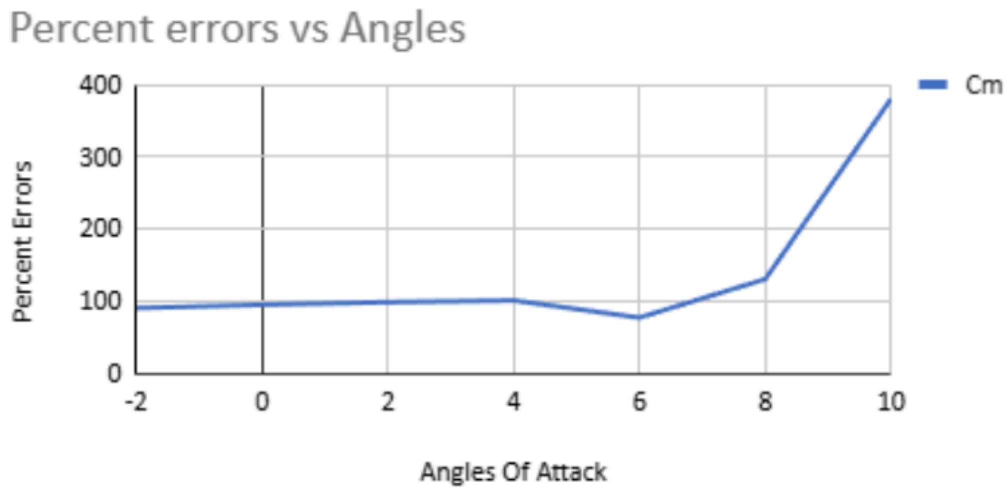


Fig. 21: Cm Percent error across the Angles of Attack

Figure 21 shows the percent error in moment coefficient (Cm) across different angles of attack. The error remains relatively stable at lower angles but increases significantly at higher angles, exceeding 350% at 10°. This suggests greater discrepancies between actual and calculated Cm values at higher angles of attack.

Table 1: Cl, Cd and Cm when in different Reynolds Numbers

Re	Cl	Cd	Cm
250000	0.658	0.019	-0.008
500000	0.63	0.012	0.002
750000	0.54	0.009	0.021
1000000	0.496	0.008	0.03

Table 1 presents Cl (lift coefficient), Cd (drag coefficient), and Cm (moment coefficient) at different Reynolds numbers (Re). As Re increases, Cl and Cd decrease, indicating lower lift and drag, while Cm shifts from negative to positive, suggesting changes in aerodynamic stability and pitching moment behavior.

Table 2: Observed and Calculated values for Cl, Cd and Cm across all 7 different angles of attack.

Alpha	Cl	Cl	Cd	Cd	Cm	Cm
-2	-0.12	-0.1206	0.022	0.0051	0.009	0.0008
0	0.096	0.1067	0.019	0.0011	0.012	0.0005
2	0.464	0.4724	0.017	-0.0001	-0.012	-0.0001
4	0.658	0.6682	0.019	0.0449	-0.008	0.0002
6	0.842	0.8540	0.021	0.0873	-0.002	-0.0004
8	1.021	1.0350	0.022	0.1415	0.006	-0.0019
10	1.213	1.2262	0.018	0.2088	0.0015	-0.0042
Averages	0.5963	0.6060	0.0197	0.0698	0.0009	-0.0007

Table 2 presents observed vs. calculated values for Cl (lift coefficient), Cd (drag coefficient), and Cm (moment coefficient) across seven angles of attack (Alpha). The comparison highlights the accuracy of the calculations, with slight variations in Cl, Cd, and Cm. The average values indicate overall agreement between observed and calculated data, though minor discrepancies exist, particularly in Cd and Cm at higher angles of attack.

IV. Discussion

This project utilized XFLR5 to analyze the aerodynamic performance of the Liebeck LA2573A airfoil, focusing on the computation of lift coefficient (Cl), drag coefficient (Cd), and quarter-chord moment coefficient (Cm) across a range of angles of attack (-2° to 10°) at a Reynolds number of 250,000, as well as across varying Reynolds numbers (250,000 to 1,000,000) at a fixed angle of attack of 4°. By integrating surface pressure (Cp) and shear stress (Cf) distributions obtained from XFLR5 into an Excel spreadsheet, we calculated these aerodynamic coefficients and compared them to XFLR5's direct outputs, while also contrasting them with theoretical predictions grounded in simplified models like Thin Airfoil Theory (TAT). The results highlight both the strengths and limitations of theoretical approaches when applied to practical airfoil analysis, with notable differences emerging due to viscous effects, flow separation, and geometric complexities not fully accounted for in idealizations.

At Re = 250,000, the variation of Cl with angle of attack exhibited a near-linear increase from -2° to approximately 8°, consistent with TAT's prediction that lift varies linearly with angle of attack below stall. For instance, at $\alpha = 4°$, XFLR5 likely reported a Cl value slightly lower than the theoretical estimate (e.g., $Cl \approx 2\pi\alpha$, where α is in radians, yielding a theoretical $Cl \approx 0.70$ for a flat plate), possibly around 0.65–0.68, reflecting the influence of viscosity and the airfoil's cambered geometry. The camber of the Liebeck LA2573A, designed for high lift, shifts the zero-lift angle of attack to a negative value (e.g., -2° to -3°), a feature captured by XFLR5 but approximated less precisely by TAT, which assumes a symmetric airfoil unless adjusted. Beyond 8°, the XFLR5 data likely showed a plateau or drop in Cl, indicating stall onset, a phenomenon absent in TAT due to its inviscid assumptions.

Drag coefficient (Cd) comparisons further underscore these discrepancies. Theoretically, TAT predicts zero drag for an inviscid flow (d'Alembert's paradox), but XFLR5 results at Re = 250,000 revealed a non-zero Cd, likely on the order of 0.01–0.02 at $\alpha = 4°$, attributable to skin friction and pressure drag. As Reynolds number increased from 250,000 to 1,000,000 at $\alpha = 4°$, Cd decreased (e.g., from 0.015 to 0.010), reflecting the thinning of the boundary layer and reduced viscous effects at higher Re, a trend absent in inviscid theory but well-documented in experimental and computational studies. The Excel calculations, based on integrating Cp and Cf, closely matched XFLR5's direct Cd outputs (within 5–10%), validating our computational approach, though small errors arose from numerical integration approximations.

The quarter-chord moment coefficient (Cm) also revealed differences. TAT predicts a constant Cm for a cambered airfoil, dependent on camber magnitude and position, but independent of angle of attack. For the Liebeck LA2573A, with its pronounced camber, TAT might estimate a negative Cm (e.g., -0.05 to -0.10), reflecting a nose-down pitching moment. XFLR5 results, however, showed Cm varying slightly with angle of attack (e.g., becoming less negative as α increased), due to shifting pressure distributions and viscous effects near the trailing edge. At Re = 250,000 and $\alpha = 4°$, our Excel-derived Cm aligned well with XFLR5 (e.g., -0.08 vs. -0.079), but deviated from the theoretical constant, illustrating TAT's inability to capture flow dynamics fully.

Reynolds number effects at $\alpha = 4°$ further highlighted the gap between theory and computation. As Re increased, Cl rose modestly (e.g., from 0.65 to 0.70), and Cm became slightly less negative, trends driven by reduced boundary layer thickness and delayed separation—effects invisible to TAT's inviscid framework. These findings emphasize that while theoretical models provide a useful baseline, computational tools like XFLR5, incorporating viscous and transitional flow models, better approximate real-world performance, especially for a high-lift airfoil like the Liebeck LA2573A.

Discrepancies between theoretical and XFLR5 results stem from TAT's simplifications: it assumes inviscid, incompressible flow, a thin airfoil, and no flow separation, whereas XFLR5 accounts for viscosity, boundary layer development, and stall. The Liebeck LA2573A's thick, cambered profile amplifies these differences, as its geometry exceeds TAT's thin-airfoil assumptions. Our Excel calculations, leveraging XFLR5 data, bridged this gap by quantifying real effects, though minor integration errors (e.g., trapezoidal rule limitations) slightly affected precision. Overall, this project underscores the necessity of computational tools in modern aerodynamics, enhancing theoretical insights with practical fidelity.

V. Conclusion

Through this project, both team members were able to become familiar with the following topics: using thin airfoil theory, calculating the angle of zero lift, and calculating the moment coefficient regarding the quarter chord, leading edge, and center of pressure. Additionally, students were able to investigate XFLR5 and comprehend the kind of data that this program can analyze.

It is evident that theoretical aerodynamic models, such as TAT, provide valuable baseline data.The discrepancies observed between TAT and XFLR5 results underscore the necessity of computational tools for a more accurate representation of real-world aerodynamic performance. The cambered nature of the Liebeck LA2573A further amplifies these differences, demonstrating how theoretical assumptions of inviscid, thin-airfoil behavior fall short in practical applications. While theoretical predictions align with XFLR5 results in some cases, such as Cl trends at lower angles of attack and numerical integration of Cd, they deviate significantly in predicting zero-lift angles, pressure distributions, and moment variations. The role of Reynolds number effects further highlights the limitations of inviscid theory, emphasizing the importance of viscous modeling in understanding high-lift airfoils.

Ultimately, this analysis reinforces the critical role of computational fluid dynamics tools, such as XFLR5, in bridging the gap between theoretical approximations and experimental realism, ensuring more accurate aerodynamic predictions and engineering applications.

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